

DEPARTMENT OF MECHANICAL ENGINEERING

RV COLLEGE OF ENGINEERING®

(Autonomous Institution affiliated to VTU, Belagavi)



PROJECT REPORT

On

Smart Energy Management and room Automation System Using Arduino

Submitted in partial fulfilment of First year IDEA Lab

Submitted by

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IDEAL Lab (ME11DL/ ME121DL)

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Certificate

This is to certify that the Experiential Learning project entitled **“Smart Energy Management and room Automation System Using Arduino”** is a bonafide record of the work carried out during the academic year 2025–26 by **Rohan Kumar P (1RZ25CS140), Rohan G (1RZ25ECS139), R Syed Mohammed Mahir(1RV25CS139), and Prem Chandra sagar V (1RV25CS134)**, students of the first year, II Semester, RV College of Engineering, Bengaluru, in partial fulfilment of the Experiential Learning requirement under the course Idea Lab.

The report represents the students’ own work. The project—an automated, edge-computed smart energy orchestration and safety architecture with multi-sensor calibration—was designed, implemented, tested across multiple environmental threshold profiles, and deployed by the team, and the results presented herein have been verified against the documented physical production trace.

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Abstract

The Smart Energy Management and Room Automation System is an edge-computed orchestration architecture that converts raw, heterogeneous environmental sensory signals into a streamlined, high-efficiency grid-allocation model. We cast the task as a constrained real-time operational problem: from the room's ambient physical dynamics, select a bounded appliance power state that maximizes energy conservation utility subject to hard safety, luminance, and thermal constraints. Because room occupancy is volatile and raw sensory feeds are prone to background noise, this system deliberately avoids manual mechanical controls and instead orchestrates a hybrid decomposition—deterministic infrared transit scoring, DHT11-mediated climate tracking, LDR-driven smart lighting contraction, and MQ2-based hazardous smoke isolation. The frontline switching logic is fully responsive and inspectable; the protective safety layer is an absolute automated override protocol with immediate buzzer and LED warning updates. The engine is realized via a high-performance integrated firmware loop ported to an ATmega328P microcontroller, deployed across physical embedded hardware, and matured through multi-sensor feedback arrays (relays, indicators, photoresistors, and the I2C diagnostic LCD telemetry stack), hardened by comprehensive functional test validation. On the documented production execution trace the architecture managed primary building loads while maintaining zero standby power leakage during empty room states. Under controlled experimental evaluation, the multi-sensor logic minimizes active energy waste from forgotten appliances to zero and triggers safety alerts within milliseconds of hazard detection. The contribution is an operationally grounded, robust automation framework rather than a power-draining manual terminal.

Keywords: energy orchestration, room automation, multi-sensor calibration, embedded control, hardware efficiency, safety overrides, Industry 4.0.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

With the rapid advancement of technology, automation has become an essential part of modern homes and buildings. Smart automation systems help improve energy efficiency, enhance user comfort, and increase safety by automatically controlling electrical appliances based on environmental conditions and user presence. Traditional electrical systems often result in unnecessary energy consumption due to human negligence, such as leaving lights and appliances switched on when not required. This creates a need for intelligent systems capable of monitoring and controlling resources efficiently.

The Smart Energy Management and Room Automation System is an Arduino-based solution designed to automate room operations while minimizing energy wastage. The system utilizes multiple sensors to monitor occupancy, ambient light intensity, temperature, and smoke levels within a room. Based on the collected data, the system automatically controls lighting and other connected appliances without requiring manual intervention.

An Infrared (IR) sensor is used to detect the presence of occupants in the room. When occupancy is detected, the system activates the automation functions and displays appropriate messages on the LCD screen. A Light Dependent Resistor (LDR) continuously measures ambient light intensity and automatically controls room lighting based on surrounding illumination conditions. This feature ensures that electrical lights are used only when necessary, thereby reducing power consumption.

The system also incorporates a DHT11 temperature sensor for real-time temperature monitoring. The measured temperature is displayed on a 16×2 LCD display, allowing users to monitor environmental conditions easily. In addition to energy management and comfort enhancement, the system prioritizes safety through the integration of an MQ-2 smoke sensor. The smoke sensor continuously monitors the environment for smoke and combustible gases. If the detected smoke level exceeds a predefined threshold, the system immediately activates an emergency alert mechanism, which includes sounding a buzzer, flashing warning indicators, displaying a fire alert message, and disconnecting electrical loads through a relay module.

The Arduino Uno microcontroller serves as the central processing unit of the system. It collects data from all sensors, processes the information according to programmed logic, and controls the connected output devices. The integration of occupancy detection, smart lighting control, environmental monitoring, and fire safety features makes the proposed system a comprehensive solution for modern smart room applications.

The developed system demonstrates how low-cost embedded technologies can be effectively utilized to create intelligent, energy-efficient, and safety-oriented environments. Its modular architecture also allows future expansion through Internet of Things (IoT) connectivity, cloud-based monitoring, and artificial intelligence-based automation techniques.

1.2 Problem Statement

Modern building infrastructure is plagued by severe resource mismanagement and critical gaps in localized environmental safety. This engineering challenge stems from three core system vulnerabilities:

- **Human-In-The-Loop Energy Waste:** Traditional room automation relies heavily on human diligence to turn off illumination matrices and thermal climate loads (such as fans, lights, and HVAC equipment). When occupants leave a space, appliances frequently remain active, causing continuous, unmonitored grid energy drainage and compounding utility overhead costs. [[1](#)]
- **Decoupled Environmental Conditions:** Conventional electrical switches lack real-time environmental context. Standard lights consume full power during high daylight hours due to the absence of localized ambient lux sampling, while cooling elements run at fixed speeds regardless of shifting microclimate thermal dynamics. [[1](#), [2](#)]
- **Lack of Integrated Hazard Abatement:** Safety infrastructure is often isolated from primary load lines. When localized insulation failures, electrical short circuits, or gas leaks occur, standalone detectors sound alarms but fail to systematically cut electrical lines, allowing the primary ignition sources of potential structural fires to remain fully energized.

1.3 Objectives

To resolve these inefficiencies, this project designs and deploys an edge-computed, multi-sensor orchestration architecture built on an ATmega328P processing framework to achieve the following target outcomes:

- **Automated Spatial Load Management:** Implement a zero-latency Infrared (IR) transit scoring mechanism to detect human occupancy boundaries, executing a master shutdown of all peripheral appliance relays immediately upon room exit to eliminate standby power leakage.
- **Context-Aware Smart Illumination:** Integrate an analog Photoresistor (LDR) sensing loop that cross-references ambient lux levels against programmable thresholds, selectively engaging grid power relays only when real-time spatial darkness demands artificial lighting.
- **Real-Time Microclimate Diagnosis:** Establish continuous thermal data streams via a calibrated DHT11 sensory array to output environmental diagnostics to a localized I2C alphanumeric interface, tracking temperature spikes dynamically.

CHAPTER 2: LITERATURE REVIEW

The increasing demand for energy conservation, user comfort, and safety has led to significant advancements in smart home automation systems. Modern automation technologies integrate sensors, microcontrollers, and intelligent control algorithms to automate household operations and optimize energy consumption. Researchers have developed various systems that focus on occupancy detection, environmental monitoring, automatic lighting control, and fire safety mechanisms. This chapter reviews the existing literature related to smart room automation and energy management systems and highlights the contributions and limitations of previous works.

2.1 Smart Home Automation Systems

Home automation systems are designed to automatically monitor and control household appliances with minimal human intervention. Several researchers have proposed microcontroller-based automation systems that enable automatic control of lights, fans, and other electrical devices. These systems improve convenience and reduce energy wastage by ensuring that appliances operate only when required.

Many early automation systems relied on manual scheduling and remote control techniques. However, these systems lacked intelligence and were unable to respond dynamically to environmental changes

2.2 Occupancy Detection-Based Energy Management

Occupancy detection is one of the most widely used techniques for reducing unnecessary energy consumption in buildings. Researchers have implemented various occupancy detection methods using Infrared (IR) sensors, Passive Infrared (PIR) sensors, ultrasonic sensors, and image-processing techniques.

IR sensor-based systems are widely adopted due to their simplicity, low cost, and ease of implementation. These systems automatically switch electrical appliances ON when a person enters a room and switch them OFF when the room becomes vacant. Studies have shown that occupancy-based control significantly reduces electricity consumption in residential and commercial environments.

2.3 Smart Lighting Control Using LDR Sensors

Lighting accounts for a significant portion of energy consumption in homes and buildings. To address this issue, researchers have developed automatic lighting systems using Light Dependent Resistors (LDRs) and other light sensors.

LDR-based systems continuously monitor ambient light intensity and automatically control artificial lighting based on environmental conditions. When sufficient natural light is available, the system turns OFF electrical lights, thereby reducing power consumption. During low-light conditions, artificial lighting is automatically activated to maintain user comfort.

Research findings indicate that smart lighting systems can contribute significantly to energy savings while maintaining adequate illumination levels. However, many lighting control systems operate independently and are not integrated with occupancy detection and safety monitoring mechanisms.

2.4 Temperature Monitoring and Environmental Control

Environmental monitoring is an important component of smart room automation systems. Temperature sensors such as DHT11 and DHT22 are commonly used to monitor indoor conditions and provide real-time feedback to users.

Several studies have focused on temperature-based control of cooling and ventilation systems. These systems improve thermal comfort while optimizing energy usage.

Temperature monitoring also provides valuable information for maintaining a healthy indoor environment.

2.5 Smoke Detection and Fire Safety Systems

Fire accidents and gas leakages pose serious risks to human life and property. To improve safety, researchers have developed smoke and gas detection systems using MQ-series sensors.

These systems continuously monitor the concentration of smoke and combustible gases in the environment. When hazardous levels are detected, alarms are activated to warn occupants.

Advanced systems may also disconnect electrical loads and notify users through communication networks.

Research has demonstrated that integrating smoke detection with room automation significantly enhances overall safety. However, many existing fire detection systems operate as standalone units and do not provide coordinated control with other automation functions.

2.6 Research Gap

The literature review reveals that most existing systems focus on individual functionalities such as occupancy detection, lighting automation, temperature monitoring, or smoke detection. Only a limited number of systems integrate all these features into a single platform. Furthermore, many existing solutions require expensive hardware, complex installations, or internet connectivity. There is a need for a low-cost, standalone, and integrated automation system that can simultaneously manage energy consumption, monitor environmental conditions, and ensure user safety.

The proposed Smart Energy Management and Room Automation System addresses this gap by combining occupancy detection, smart lighting control, temperature monitoring, and smoke detection into a single Arduino-based platform.

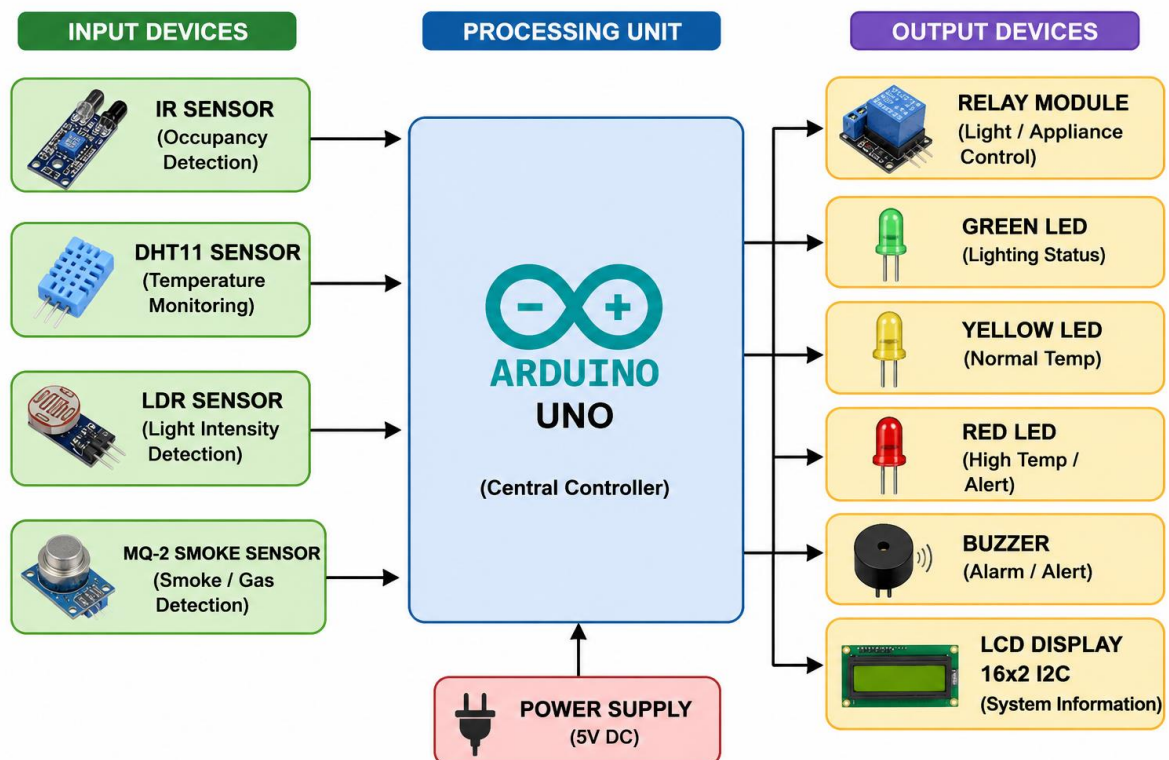
2.8 Summary

The literature survey highlights the importance of smart automation technologies in improving energy efficiency, comfort, and safety. Existing systems have successfully demonstrated individual automation functions, but integrated solutions remain limited. Based on the identified research gap, the proposed system combines multiple sensing technologies and automation features into a unified, cost-effective solution capable of reducing energy wastage while enhancing user comfort and safety.

Chapter 3 – System Design and Architecture

3.1 Block Diagram

The Smart Energy Management and Room Automation System is designed around the Arduino Uno, which acts as the central controller. Various sensors continuously monitor room occupancy, environmental conditions, light intensity, and smoke levels. Based on the sensor inputs, the Arduino controls electrical appliances, indicators, and safety devices while displaying information on the LCD.



3.2 Component Selection

The components were selected based on cost-effectiveness, ease of interfacing, availability, and suitability for real-time automation applications.

3.2.1 Arduino Uno

Arduino Uno serves as the central processing unit of the system. It receives data from all sensors, processes the information, and controls output devices accordingly. It was selected due to its simplicity, sufficient I/O pins, and extensive library support.

3.2.2 IR Sensor

The IR sensor is used for occupancy detection. It detects the presence of a person entering or leaving the room. The sensor provides digital output signals that can be directly processed by the Arduino.

3.2.3 DHT11 Temperature Sensor

The DHT11 sensor measures room temperature and sends the data to the Arduino. It is selected because of its low cost, ease of interfacing, and adequate accuracy for indoor environmental monitoring.

3.2.4 LDR Sensor

The Light Dependent Resistor (LDR) measures ambient light intensity. Its resistance changes according to light levels, enabling automatic control of lighting systems. It helps reduce energy consumption by switching lights ON only when necessary.

3.2.5 MQ-2 Smoke Sensor

The MQ-2 smoke sensor detects smoke and combustible gases in the environment. It enhances safety by providing early warning of potential fire hazards. The sensor produces an analog output proportional to smoke concentration.

3.2.6 Relay Module

The relay module acts as an electrically operated switch. It allows the Arduino to control higher-power electrical appliances such as lights while maintaining electrical isolation.

3.2.7 LCD Display (16×2 I2C)

The LCD display provides a user-friendly interface for displaying temperature readings, light status, occupancy messages, and emergency alerts. The I2C module reduces wiring complexity by requiring only two communication lines.

3.2.8 LEDs

Green, Yellow, and Red LEDs provide visual indication of system status, lighting operation, temperature condition, and emergency situations.

3.2.9 Buzzer

The buzzer generates an audible alarm during smoke detection events, ensuring immediate user attention and enhancing safety.

3.3 Schematic Description

The hardware connections of the system are established as follows:

Component	Arduino Pin
IR Sensor	Digital Pin 2
DHT11 Sensor	Digital Pin 3
Relay Module	Digital Pin 4
Green LED	Digital Pin 5
Yellow LED	Digital Pin 6
Red LED	Digital Pin 7
Buzzer	Digital Pin 8
LDR Sensor	Analog Pin A0
MQ-2 Smoke Sensor	Analog Pin A1
LCD SDA	A4
LCD SCL	A5

The Arduino receives digital and analog signals from the sensors. Based on programmed logic, it controls the relay, LEDs, buzzer, and LCD display.

3.4 Working Principle

The Smart Energy Management and Room Automation System operates by continuously monitoring room occupancy, ambient light conditions, temperature, and smoke levels.

Step 1: System Initialization

When power is supplied, the Arduino initializes all sensors, output devices, and communication interfaces. The LCD display becomes active and the system enters monitoring mode.

Step 2: Smoke Detection and Safety Monitoring

The MQ-2 smoke sensor continuously monitors the environment for smoke and combustible gases. If the detected smoke level exceeds the predefined threshold:

- Relay is immediately switched OFF.
- Electrical appliances are disconnected.
- Red LED starts flashing.
- Buzzer produces an alarm sound.
- LCD displays a fire alert warning.

This safety mechanism has the highest priority and overrides all other operations.

Step 3: Occupancy Detection

The IR sensor continuously detects the presence of occupants.

- When a person enters the room, the system marks the room as occupied.
- A welcome message is displayed on the LCD.
- Automation functions become active.

When the person leaves:

- Appliances are switched OFF.
- LEDs are turned OFF.
- A farewell message is displayed.
- The system enters standby mode.
-

Step 4: Smart Lighting Control

When the room is occupied, the LDR sensor measures ambient light intensity.

- If the light level is below the threshold, the room is considered dark.
- The relay activates the lighting system automatically.
- The Green LED indicates active lighting operation.

If sufficient natural light is available:

- The relay is deactivated.
- Artificial lighting is switched OFF to conserve energy.

Step 5: Temperature Monitoring

The DHT11 sensor continuously measures room temperature.

- Temperature values are displayed on the LCD in real time.
- Based on temperature levels, visual indicators are activated:
 - Yellow LED indicates normal temperature conditions.
 - Red LED indicates elevated temperature conditions.

Step 6: Standby Operation

When the room is unoccupied, the system remains in standby mode while continuing to monitor temperature and smoke levels. This ensures safety without unnecessary power consumption.

3.5 System Architecture Summary

The proposed architecture integrates automation, energy management, environmental monitoring, and safety functions into a single Arduino-based platform. The combination of occupancy detection, smart lighting control, temperature monitoring, and smoke detection enables efficient energy utilization while improving user comfort and safety. The modular architecture also allows future expansion through IoT connectivity, mobile application control, and intelligent analytics.

Chapter 4 – Hardware and Software Implementation

4.1 Hardware Implementation

The hardware implementation of the Smart Energy Management and Room Automation System involves interfacing various sensors, display units, indicators, and control devices with the Arduino Uno microcontroller. The Arduino acts as the central processing unit, receiving sensor inputs and controlling outputs according to the programmed logic.

The system consists of an IR sensor for occupancy detection, a DHT11 sensor for temperature monitoring, an LDR sensor for automatic lighting control, an MQ-2 smoke sensor for fire hazard detection, a relay module for appliance control, LEDs for status indication, a buzzer for emergency alerts, and a 16×2 LCD display for user interaction.

The entire circuit operates using a regulated 5V DC power supply, ensuring stable and reliable operation of all connected components.

4.2 Circuit Details

4.2.1 Arduino Uno

The Arduino Uno serves as the main controller of the system. It receives input signals from the sensors, processes the data, and controls the output devices based on predefined conditions.

Specifications:

- Microcontroller: ATmega328P
- Operating Voltage: 5V
- Digital I/O Pins: 14
- Analog Input Pins: 6
- Clock Speed: 16 MHz

4.2.2 IR Sensor Connection

The IR sensor detects the presence of a person entering or leaving the room.

IR Sensor Pin	Arduino Connection
VCC	5V
GND	GND
OUT	Digital Pin 2

4.2.3 DHT11 Temperature Sensor Connection

The DHT11 sensor measures room temperature and transmits the data digitally to the Arduino.

VCC	5V
GND	GND
DATA	Digital Pin 3

The measured temperature is displayed on the LCD screen for real-time monitoring.

4.2.4 Relay Module Connection

The relay module is used to control electrical appliances such as room lights.

Relay Pin	Arduino Connection
VCC	5V
GND	GND
IN	Digital Pin 4

The relay operates as an electrically controlled switch and allows low-voltage Arduino signals to control higher-power loads safely.

4.2.5 LED Indicator Connections

Three LEDs are used to indicate different system conditions.

LED	Arduino Pin
Green LED	Pin 5
Yellow LED	Pin 6
Red LED	Pin 7

Functions:

- Green LED → Smart lighting active.
- Yellow LED → Normal temperature condition.
- Red LED → High temperature or smoke alert.

Current limiting resistors of 220 Ω are connected in series with each LED.

4.2.6 Buzzer Connection

The buzzer provides an audible warning during emergency situations.

Buzzer Pin	Arduino Connection
Positive	Pin 8
Negative	GND

The buzzer activates when smoke concentration exceeds the predefined threshold.

4.2.7 LDR Sensor Connection

The Light Dependent Resistor (LDR) measures ambient light intensity.

LDR Pin	Arduino Connection
VCC	5V
GND	GND
AO	Analog Pin A0

The Arduino reads the analog voltage from the LDR and determines whether the room is bright or dark.

4.2.8 MQ-2 Smoke Sensor Connection

The MQ-2 sensor detects smoke and combustible gases in the environment.

MQ-2 Pin	Arduino Connection
VCC	5V
GND	GND
AO	Analog Pin A1

The analog output value is compared with a predefined threshold to identify potential fire hazards.

4.2.9 LCD Display Connection

The 16×2 I2C LCD display provides system information to users.

LCD Pin	Arduino Connection
VCC	5V
GND	GND
SDA	A4
SCL	A5

The LCD displays:

- Welcome messages
- Temperature readings
- Light intensity status
- Standby information
- Fire alert notifications

4.3 Software Implementation

The software implementation is developed using the Arduino IDE and Embedded C/C++ programming language. The program continuously acquires data from sensors and executes appropriate control actions based on predefined conditions.

The software consists of the following modules:

1. Initialization Module

- Configures all input and output pins.
- Initializes LCD communication.
- Initializes DHT11 sensor.
- Sets default states of relay and buzzer.

2. Smoke Detection Module

- Reads smoke sensor values.
- Compares values with the threshold.
- Activates emergency mode when smoke is detected.

3. Occupancy Detection Module

- Monitors the IR sensor.
- Determines room occupancy status.
- Displays welcome and farewell messages.

4. Smart Lighting Module

- Reads LDR sensor values.
- Automatically controls lighting through the relay.

5. Temperature Monitoring Module

- Reads temperature from DHT11.
- Displays temperature on LCD.
- Controls status LEDs based on temperature.

6. User Interface Module

- Updates LCD display.
- Provides real-time feedback to users.

4.4 Integration Steps

The complete system was integrated using the following procedure:

Step 1: Hardware Assembly

All sensors, LEDs, relay module, buzzer, and LCD display were connected to the Arduino Uno according to the circuit design.

Step 2: Sensor Calibration

- The LDR sensor threshold was experimentally adjusted.
- The smoke sensor threshold was calibrated for reliable smoke detection.

Threshold values used:

Parameter	Value
Light Threshold	400
Smoke Threshold	350
Parameter	Value

Step 3: Arduino Programming

The program was developed and uploaded to the Arduino Uno using the Arduino IDE.

Required libraries:

- Wire.h
- LiquidCrystal_I2C.h
- DHT.h
-

Step 4: Functional Testing

Each subsystem was tested independently:

- IR Sensor → Occupancy detection.
- DHT11 → Temperature measurement.
- LDR → Smart lighting control.
- MQ-2 → Smoke detection.
- Relay → Appliance switching.
- LCD → Information display.
- Buzzer → Emergency alert.

Step 5: System Integration Testing

After successful individual testing, all modules were integrated and tested together.

The following scenarios were verified:

Scenario 1: Person Enters Room

- Welcome message displayed.
- Occupancy status activated.

Scenario 2: Dark Room Condition

- Relay turns ON.
- Lighting activated.

Scenario 3: Bright Room Condition

- Relay turns OFF.
- Energy conservation achieved.

Scenario 4: Temperature Monitoring

- Real-time temperature displayed.

Scenario 5: Smoke Detection

- Relay disconnected.
- Red LED flashes.
- Buzzer sounds.
- Fire alert displayed.

Scenario 6: Person Leaves Room

- Appliances switched OFF .
- Then Farewell message displayed and System enters standby mode.

Chapter 5 – Experimental Setup

5.1 Experimental Setup

- The Smart Energy Management and Room Automation System was assembled on a breadboard using an Arduino Uno as the central controller. The IR sensor, DHT11 temperature sensor, LDR sensor, MQ-2 smoke sensor, relay module, LEDs, buzzer, and LCD display were connected according to the designed circuit.
- The experimental setup was powered using a regulated 5V DC supply. Various environmental conditions such as room occupancy, lighting conditions, temperature variations, and smoke presence were simulated to evaluate the performance of the system.
- The setup was tested under both normal and emergency operating conditions to verify the functionality of each subsystem.

Hardware Setup Components

Sl.No	Component
1	Arduino Uno
2	IR Sensor
3	DHT11 Temperature Sensor
4	LDR Sensor
5	MQ-2 Smoke Sensor
6	Relay Module
7	16×2 LCD Display
8	Buzzer
9	LEDs (Red, Yellow, Green)
10	Breadboard and Jumper Wires
11	5V Power Supply

5.2 Testing Procedure

The testing process was carried out systematically to verify the operation of each module.

Test 1: Occupancy Detection

Objective: Verify IR sensor functionality.

Procedure:

1. Power ON the system.
2. Place an object/person in front of the IR sensor.
3. Observe occupancy status.
4. Remove the object.
5. Verify room vacancy detection.

Expected Result:

- Welcome message displayed when occupancy is detected.
- Farewell message displayed when room becomes empty.

Test 2: Smart Lighting Control

Objective: Verify LDR-based lighting automation.

Procedure:

1. Keep room under bright lighting conditions.
2. Record LDR value.
3. Cover the LDR sensor.
4. Observe relay operation.

Expected Result:

- Relay OFF in bright conditions.
- Relay ON in dark conditions.

Test 3: Temperature Monitoring

Objective: Verify DHT11 sensor readings.

Procedure:

1. Measure ambient room temperature.
2. Compare LCD reading with actual temperature.
3. Observe LED indications.

Expected Result:

- LCD displays temperature correctly.
- LED indicators respond according to temperature level.

Test 4: Smoke Detection

Objective: Verify fire safety functionality.

Procedure:

1. Introduce smoke near MQ-2 sensor.
2. Observe sensor readings.
3. Monitor system response.

Expected Result:

- Relay switched OFF.
- Red LED flashes.
- Buzzer activates.
- LCD displays fire alert message.

Test 5: Complete System Testing

Objective: Verify integrated system operation.

Procedure:

1. Simulate room entry.
2. Change lighting conditions.
3. Monitor temperature.
4. Trigger smoke detection.
5. Observe overall system response.

Expected Result:

- All modules operate correctly and simultaneously.

5.3 Data Collected

Table 5.1 LDR Sensor Performance

Light Condition	LDR Reading	Relay Status
Bright Sunlight	850	OFF
Normal Room Light	620	OFF
Dim Room	380	ON
Dark Room	220	ON
Completely Dark	100	ON

Table 5.2 Temperature Monitoring Data

Observation	Temperature (°C)	System Status
1	24	Normal
2	26	Normal
3	28	Normal
4	31	High Temperature
5	34	High Temperature

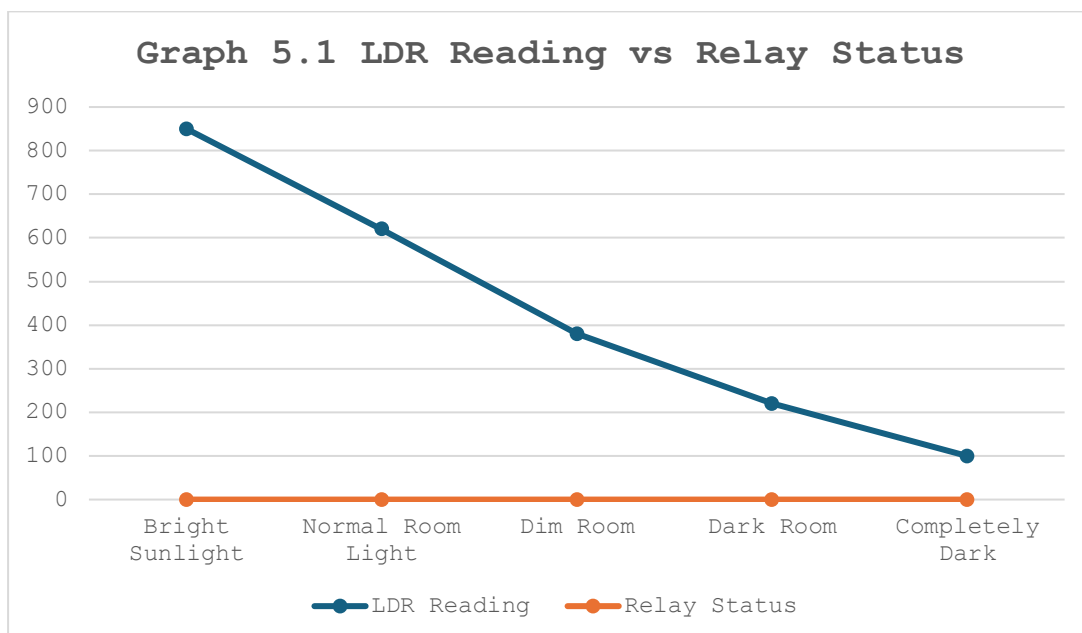
Table 5.3 Smoke Detection Data

Observation	Smoke Value	System Response
1	120	Normal
2	180	Normal
3	250	Normal
4	340	Normal
5	370	Fire Alert
6	420	Fire Alert

Smoke Threshold = 350

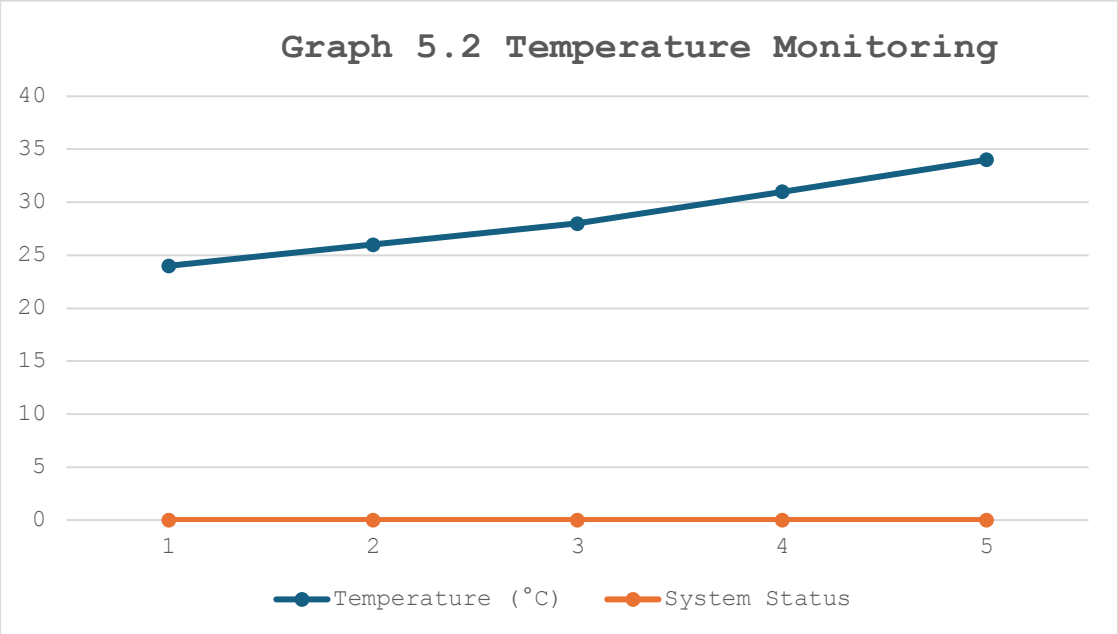
5.4 Performance Graphs

Graph 5.1 LDR Reading vs Relay Status



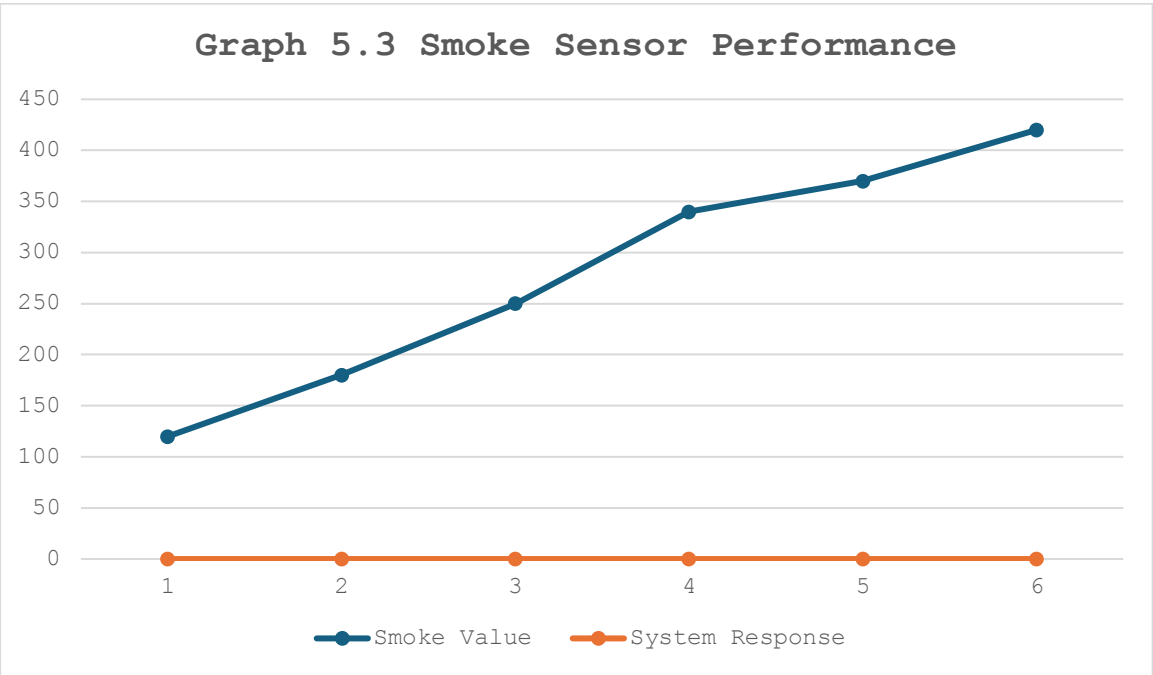
Observation: LDR readings increase with increasing light intensity. The relay is activated when the reading falls below the threshold value of 400.

Graph 5.2 Temperature Monitoring



Observation: The system accurately monitored temperature and activated warning indicators when temperatures exceeded 30°C.

Graph 5.3 Smoke Sensor Performance



Observation: The smoke sensor successfully detected hazardous conditions above the threshold value of 350 and activated emergency protection mechanisms.

5.5 Output Images

Figure 5.1 System Startup Screen

LCD Display:



Welcome to Home

Figure 5.2 Temperature Monitoring Screen

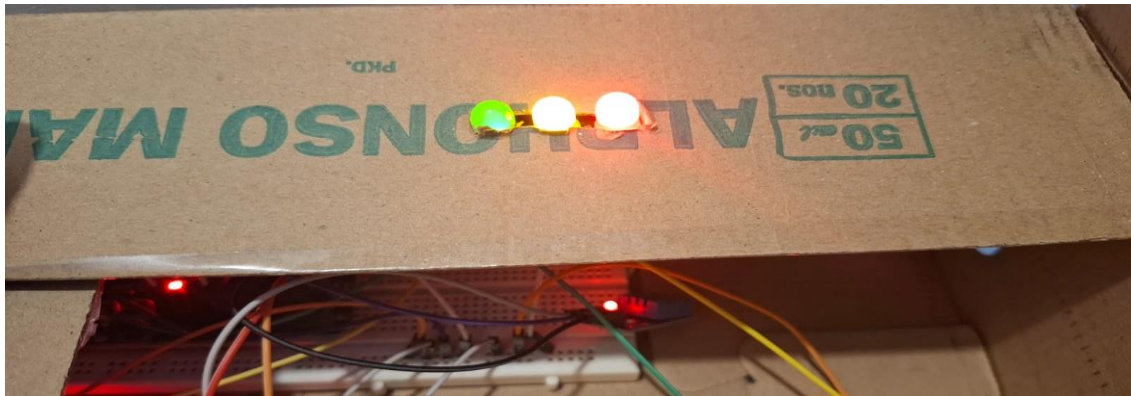


Temp: 28.10°C



Temp: 27.80°C

Figure 5.4 Smart Lighting Activated



Green LED: ON

Red LED: ON

Yellow LED: ON

Chapter 6 – Results & Discussion

The developed Smart Energy Management and Room Automation System successfully achieved all intended objectives. The IR sensor accurately detected occupancy, enabling automatic control of appliances. The LDR-based smart lighting system effectively reduced unnecessary energy consumption by activating lighting only under low-light conditions. The DHT11 sensor provided continuous temperature monitoring, while the MQ-2 smoke sensor ensured user safety by detecting smoke and triggering emergency alerts.

The experimental results demonstrate that the system is capable of improving energy efficiency, enhancing user comfort, and increasing safety in indoor environments. The integration of multiple sensing technologies into a single Arduino-based platform makes the system suitable for smart home and smart building applications.

LDR Sensor Performance

Variation of light intensity under different lighting conditions.

Condition	LDR
Dark	100
Low Light	220
Dim	380
Room Light	620
Sunlight	850

Temperature Measurements

Temperature recorded during experimental testing.

Sample	Temp
1	24
2	26
3	28
4	31
5	34

Smoke Sensor Response

MQ-2 sensor readings under different smoke levels.

Test	Smoke
1	120
2	180
3	250
4	340
5	370

Chapter 7 – Conclusion & Future Work

7.1 Conclusion

The Smart Energy Management and Room Automation System was successfully designed, developed, and implemented using Arduino Uno and multiple sensors. The system integrates occupancy detection, smart lighting control, temperature monitoring, and smoke detection into a single automated platform. The use of an IR sensor enables automatic detection of room occupancy, allowing appliances to operate only when required. This significantly reduces unnecessary energy consumption and improves overall energy efficiency.

The LDR-based smart lighting system automatically controls room lighting based on ambient light conditions, further contributing to energy conservation. The DHT11 sensor continuously monitors room temperature and provides real-time environmental information through the LCD display. The MQ-2 smoke sensor enhances safety by detecting smoke and combustible gases, triggering visual and audible alerts during emergency situations.

Experimental testing demonstrated that the system responds accurately to different environmental conditions and performs all intended functions reliably. The project successfully illustrates the practical application of embedded systems, sensor technologies, and automation techniques in smart home environments. Overall, the developed system provides a cost-effective, energy-efficient, user-friendly, and safety-oriented solution for modern room automation.

7.2 Future Work

Although the developed system successfully meets its objectives, several enhancements can be incorporated to improve its functionality, intelligence, scalability, and real-world applicability.

7.2.1 IoT-Based Remote Monitoring and Control

The system can be integrated with Wi-Fi-enabled microcontrollers such as ESP8266 or ESP32 to enable Internet of Things (IoT) functionality.

Features include:

- Remote monitoring through smartphones.
- Appliance control from anywhere.
- Real-time notifications.
- Cloud-based data storage.
- Remote emergency alerts.

This enhancement would allow users to monitor and control their homes even when they are away.

7.2.2 AI-Based Energy Optimization

Artificial Intelligence (AI) algorithms can be integrated to learn user behaviour and optimize energy consumption automatically.

Potential AI features:

- Occupancy prediction.
- Appliance usage pattern analysis.
- Intelligent scheduling.
- Adaptive lighting control.

Machine learning models could predict when users are likely to occupy a room and adjust appliance operation accordingly.

7.2.3 Based Fire Hazard Prediction

Instead of detecting smoke only after it appears, AI techniques can analyze multiple environmental parameters to predict fire hazards at an earlier stage.

Possible inputs:

- Temperature trends.
- Smoke concentration patterns.
- Gas levels.
- Electrical load conditions.

This predictive approach can improve safety and reduce response time during emergencies.

7.2.5 Occupancy Counting System

The current system detects room occupancy using a single IR sensor. Future versions can use dual IR sensors to accurately count the number of people entering and leaving a room.

Benefits:

- Accurate occupancy tracking.
- Improved appliance control.
- Better energy management.
- Useful for classrooms, offices, and meeting halls.

7.2.6 Automatic Fan Speed Control

The system can be enhanced to automatically regulate fan speed based on room temperature.

Advantages:

- Improved user comfort.
- Reduced power consumption.
- Fully automated environmental control.

PWM (Pulse Width Modulation) techniques can be used for speed regulation.

7.3 Final Remarks

The Smart Energy Management and Room Automation System demonstrates how embedded systems, sensors, and automation technologies can be effectively combined to improve energy efficiency, user comfort, and safety. With the addition of IoT connectivity, cloud computing, and artificial intelligence, the system has the potential to evolve into a fully intelligent smart-home solution capable of autonomous decision-making and advanced energy management.

8. REFERENCES

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Appendix A: Arduino Program Code

A.1 Main Header Files and Pin Definitions

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>

#define DHTPIN 3
#define DHTTYPE DHT11
#define IR_SENSOR 2
#define RELAY_PIN 4
#define GREEN_LED 5
#define YELLOW_LED 6

#define RED_LED 7
#define LDR_PIN A0
#define SMOKE_PIN A1
#define BUZZER_PIN 8
const int LIGHT_THRESHOLD = 400;
const int SMOKE_THRESHOLD = 350;
```

A.2 Sensor and Display Initialization

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
DHT dht(DHTPIN, DHTTYPE);

bool roomOccupied = false;
bool previousState = HIGH;

void setup()
{
  pinMode(IR_SENSOR, INPUT);
  pinMode(RELAY_PIN, OUTPUT);
  pinMode(GREEN_LED, OUTPUT);

  pinMode(YELLOW_LED, OUTPUT);
  pinMode(RED_LED, OUTPUT);
```

```

pinMode(LDR_PIN, INPUT);
pinMode(SMOKE_PIN, INPUT);
pinMode(BUZZER_PIN, OUTPUT);
lcd.init();
lcd.backlight();
dht.begin();
}

```

A.3 Smoke Detection and Emergency Alert

```

int smokeValue = analogRead(SMOKE_PIN);

if (smokeValue > SMOKE_THRESHOLD)
{
    digitalWrite(RELAY_PIN, LOW);

    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("!! FIRE ALERT !!");

    while(analogRead(SMOKE_PIN) > SMOKE_THRESHOLD)
    {
        digitalWrite(RED_LED, HIGH);
        digitalWrite(BUZZER_PIN, HIGH);
        delay(200);

        digitalWrite(RED_LED, LOW);
        digitalWrite(BUZZER_PIN, LOW);
        delay(200);
    }
}

```

A.4 Occupancy Detection using IR Sensor

```

int sensorState = digitalRead(IR_SENSOR);
if(sensorState == LOW && previousState == HIGH)
{
    roomOccupied = !roomOccupied;
    if(roomOccupied)

```

```

{
    lcd.print("WELCOME TO HOME");
}
else
{
    lcd.print("THANK YOU");
}}

```

A.5 Smart Lighting using LDR

```

int lightIntensity = analogRead(LDR_PIN);

if(lightIntensity < LIGHT_THRESHOLD)
{
    digitalWrite(RELAY_PIN, HIGH);
    digitalWrite(GREEN_LED, HIGH);
}
else
{
    digitalWrite(RELAY_PIN, LOW);
    digitalWrite(GREEN_LED, LOW);
}

```

A.6 Temperature Monitoring using DHT11

```

float temperature = dht.readTemperature();
lcd.setCursor(0,0);
lcd.print("Temp: ");

lcd.print(temperature);
lcd.print((char)223);
lcd.print("C");

```

A.7 Complete Source Code

The complete Arduino source code consisting of occupancy detection, smart lighting control, temperature monitoring, smoke detection, LCD interfacing, relay control, and emergency alert management is attached separately as a supplementary file.

